



Project Management Overview

This note combines key concepts from four PDFs, covering the fundamentals of project management, planning and scheduling practices, a placeholder document, and an introduction to the PERT scheduling technique.



Content Sources

- Section 1: Provides a comprehensive overview of **Project Management** fundamentals, including project definition, temporary nature, and unique deliverables, from a 65-page PDF document. ([source](#))
- Section 2: Outlines the **Planning & Scheduling** process, baseline creation, and monitoring activities, from a 44-page PDF document. ([source](#))
- Section 3: Contains only blank white pages with **No Text**, from an 11-page PDF document. ([source](#))
- Section 4: Explains the **PERT** technique, its probabilistic duration handling, and relation to CPM, from a 13-page PDF document. ([source](#))

Section 1: Introduction to Project Management Fundamentals

This section covers foundational project management concepts, definitions, and frameworks from a 65-page PDF document ([source](#))

1.1 What Is a Project?

Project: A *temporary* endeavour undertaken to create a **unique** product, service, or result (PMBOK Guide definition)

Characteristic	Explanation
Temporary	Has defined start and end; ends when objectives are met or when objectives

	cannot be met. Duration is finite but not necessarily short.
Unique Deliverables	Creates something distinct that doesn't exist before
Progressive Elaboration	Scope described broadly early, refined incrementally as understanding deepens. Not scope creep.

Types of Unique Deliverables:

- **Product/Artifact** – quantifiable, may be end item or component
- **Capability** – ability to perform a service (e.g., business functions)
- **Result** – outcomes, documents, knowledge (e.g., research findings)

Shared Characteristics with Operations:

- Performed by people
- Constrained by limited resources
- Planned, executed, and controlled

1.2 Core Project Characteristics

The Iron Triangle (Constraints):

Constraint	Description
Cost	Budget and financial resources
Time	Schedule and deadlines
Scope	Work to be performed
Quality	Additional dimension – how well the deliverables meet requirements

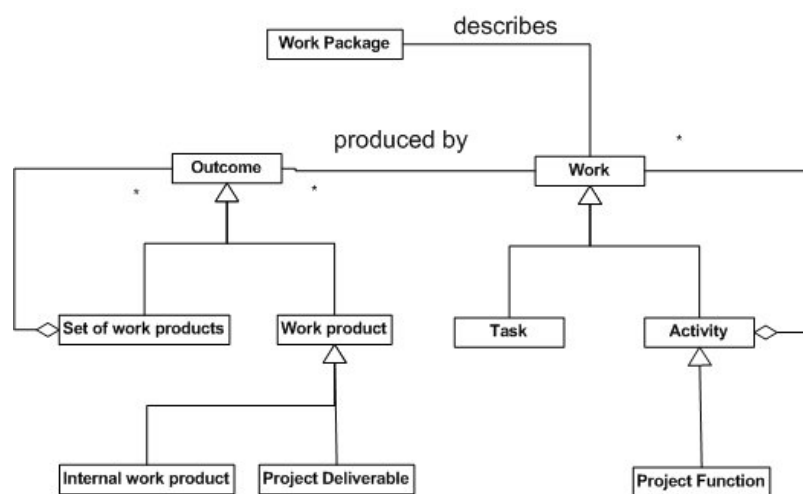
Trade-offs among these factors shape decisions throughout the project lifecycle

1.3 Project Components (Alternative View)

Component	Description
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Outcome	Work product(s) delivered to customer; also called deliverables
Work	Effort required to achieve outcome. Broken into: work packages → tasks (smallest manageable unit) → activities (can be grouped recursively)
Schedule	Timeline mapping of work units, showing start, duration, end, and dependencies
Resource	People (participants), funds, equipment, facilities. Participants organized into organization and assume roles responsible for work packages

Visualizing Relationships:



The flowchart illustrates how work packages produce outcomes, how tasks/activities generate work products, and how these elements interrelate within a project.

1.4 What Is Project Management? 🎯

Project Management: The application of **knowledge, skills, tools, and techniques** to project activities in order to meet project requirements

Five Process Groups (integrated throughout project):

Process Group	Purpose
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1. Initiating	Define and authorize project or phase
2. Planning	Establish scope, refine objectives, define actions
3. Executing	Complete work defined in plan
4. Monitoring & Controlling	Track, review, regulate progress and performance
5. Closing	Finalize all activities, formally close project or phase

Functional Viewpoint (Alternative):

Project Management as *guiding process* with four basic activities:

Activity	Description
Planning	Define results, tasks, schedule, resources
Organizing	Set up project structure, assign roles & responsibilities
Controlling	Detect deviations, adjust resources/schedule/outcome as needed
Terminating	Deliver system, obtain acceptance, capture lessons learned, update templates

1.5 Project Manager's Role

Primary Responsibility: Achieve project objectives

Key Duties:

- Identify requirements
- Set clear objectives
- Balance **quality/scope/time/cost**
- Adapt specifications for stakeholder expectations

Broader Use of Term: "Project Management" can denote organizational approach treating many activities (including some operations) as projects — "Management by projects"

1.6 PMBOK Guide — Knowledge Areas & Core Processes

Knowledge Area	Core Processes (selected)
Project Integration Management	Develop Project Charter, Preliminary Scope Statement, Project Management Plan; Direct & Manage Execution; Monitor & Control Work; Integrated Change Control; Close Project
Project Scope Management	Scope Planning, Scope Definition, Create WBS , Scope Verification, Scope Control
Project Time Management	Activity Definition, Sequencing, Resource Estimating, Duration Estimating, Schedule Development, Schedule Control
Project Cost Management	Cost Estimating, Cost Budgeting, Cost Control
Project Quality Management	Quality Planning, Perform Quality Assurance, Perform Quality Control
Project Human Resource Management	HR Planning, Acquire Project Team, Develop Project Team, Manage Project Team
Project Communications Management	Communications Planning, Information Distribution, Performance Reporting, Manage Stakeholders
Project Risk Management	Risk Planning, Identification, Qualitative & Quantitative Analysis, Response Planning, Monitoring & Control
Project Procurement Management	Plan Purchases & Acquisitions, Plan Contracting, Request Seller Responses, Select Sellers, Contract Administration, Contract Closure

1.7 Organizational Structures

General Concepts

- **Organization** consists of **organizational units** (participants, teams, divisions, departments)

- **Project-based organization** — Participants assigned to only one project at a time; projects created on demand and terminated when outcomes delivered

Functional (Line) Organization

- Participants grouped by specialty into **departments** (e.g., Accounting, Finance, Marketing, Production, MIS)
- Projects flow through **pipeline** of departments; each department permanent part of organization

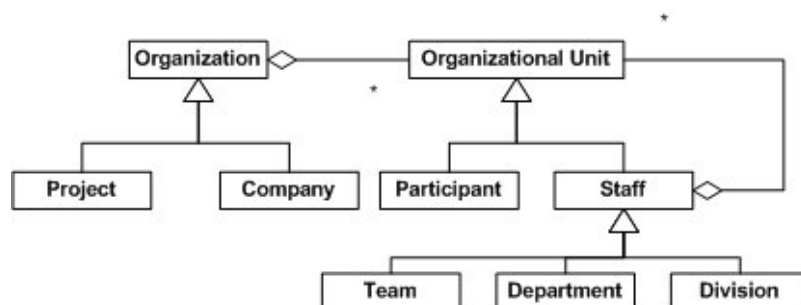
Geographic, Product, Customer, Process Differentiation

Type	Description
Geographic	Teams located in different cities/regions (e.g., St Louis, Baltimore)
Product	Separate units for distinct product families (e.g., Tobacco, Health Products, Tankers)
Customer	Units oriented to specific client types (e.g., Military Sales, Covert Sales)
Process	Functional units aligned to phases of system lifecycle (Analysis, Programming, Training)

Matrix Organization

- **Hybrid** of functional and project-based structures
- Participants split time between department duties and project assignments
- **Dual reporting** — Each team member reports to both **department head** and **project manager**, violating Fayol's "unity of command" but enabling rapid response and skill-based staffing

Visual Overview:



The diagram shows the hierarchy from "Organization" → "Organizational Unit" → branches such as Project, Company, Participant, Staff, Team, Department, Division.

1.8 Management Theories

Henri Fayol (Classical Management)

Five Functions of Management:

Function	Description
Forecast & Plan	Anticipate future conditions and devise actions (Prevoyance)
Organize	Build material and human structures
Command	Direct personnel activity
Coordinate	Harmonize all activities
Control	Ensure conformity with rules and commands

14 Principles of Management:

#	Principle	Description
1	Specialization of Labor	Skill development and method improvement
2	Authority	Right to give orders, power to obtain obedience
3	Discipline	Adherence to rules
4	Unity of Command	One boss per employee
5	Unity of Direction	Single plan guiding all efforts
6	Subordination of Individual Interests	Work interests dominate personal ones
7	Remuneration	Fair payment

8	Centralization	Decision-making concentration (top-down)
9	Scalar Chain	Formal line of authority
10	Order	Proper placement of materials and personnel
11	Equity	Fair treatment
12	Personnel Tenure	Job stability, long-term employment
13	Initiative	Encourage planning and execution
14	Esprit de Corps	Team spirit and cohesion

Henry Mintzberg (Behavioural Management)

Key Quote: "There is surprisingly little study of what managers actually do."

Managerial Roles (10) — Grouped into three clusters:

Cluster	Roles	Description
Interpersonal	Figurehead, Leader, Liaison	People-focused interactions
Informational (~40% of time)	Monitor, Disseminator, Spokesperson	Information gathering and sharing
Decisional	Entrepreneur, Disturbance Handler, Resource Allocator, Negotiator	Action-oriented choices

Gestalt View: The ten roles are inseparable; together they form a whole greater than the sum of parts.

Role Highlights (Relevant to Project Managers):

Role	Key Activities
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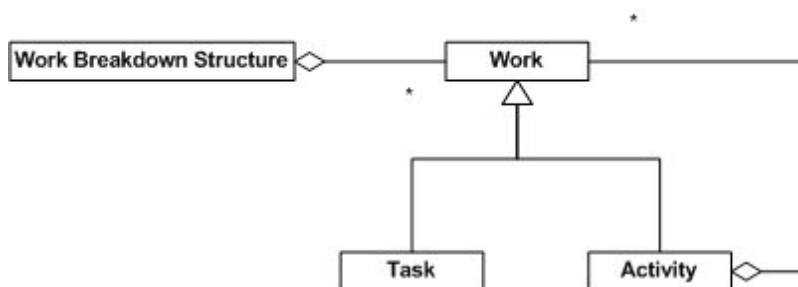
Figurehead	Ceremonial duties (lunches, attending events)
Leader	Direct and indirect leadership; motivate team via influence rather than authority
Liaison	Build external contacts; gather information beyond vertical chain
Monitor	Scan environment for soft information (rumors, speculation)
Disseminator	Share information internally; essential for decision making
Spokesperson	Communicate outward to senior managers, clients, stakeholders
Entrepreneur	Initiate new projects or improvements based on ideas spotted in monitor role
Disturbance Handler	React to unexpected changes (scope shifts, accidents, conflicts)
Resource Allocator	Decide who gets what (including manager's own time)

1.9 Project Planning Foundations

Work Breakdown Structure (WBS)

Definition: Hierarchical decomposition of the total scope into **tasks** (or work packages)

- **Purpose:** Provides simple, aggregate view of all work to be performed



The diagram shows top-level "Work Breakdown Structure" branching into "Work," then into "Task" and "Activity," visualizing hierarchical breakdown.

Task Model (Network Diagram)

Element	Description
Temporal Dependencies	Certain tasks must precede others
Duration	Estimated time required given assigned participants
Critical Path	Longest path through network; tasks have zero slack — delays affect overall finish date
Slack (Float)	Maximum postponement possible for non-critical task without delaying project

Skill (Competency) Matrix

After WBS, manager matches **participants** to **tasks** using skills matrix:

Symbol	Meaning
● (filled circle)	Primary skill — qualified to lead the task
○ (empty circle)	Secondary skill — qualified to participate
△ (triangle)	Interest — wants involvement but lacks skill

Example Matrix:

Tasks \ Participant	John	Mary	Kofi	Robert
Control Design	●	○	●	—
Database Design	●	●	○	—
User Interface Design	△	●	○	—

Configuration Management	△	●	–	○
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Interpretation: Primary-skill owners become **task leads**, secondary-skill owners assist, interested participants may be coached.

1.10 Work Products, Work Packages & Roles

Element	Description
Work Product	Any tangible

Putting it All Together

Integrated Foundations of Projects

- A **project** is a *temporary* effort producing a **unique** outcome—product, capability, or result.

“*Temporary*” means it has a defined start and end; “*unique*” distinguishes it from routine operations.

- Core **constraints** (the Iron Triangle) are **Cost, Time, Scope**, with **Quality** as a cross-cutting dimension. Trade-offs among these shape every decision.
- **Project management** applies *knowledge, skills, tools, and techniques* across the five process groups: **Initiating, Planning, Executing, Monitoring & Controlling, Closing**.
- The **PMBOK** knowledge areas (Integration, Scope, Schedule, Cost, Quality, Resources, Communications, Risk, Procurement) provide a structured checklist for each phase.

Planning & Scheduling Mechanics

- **Work Breakdown Structure (WBS)** hierarchically decomposes the total scope into **work packages**, then **tasks** and **activities**—the backbone for estimating, scheduling, and assigning resources.
- Key planning artifacts:

Artifact	Purpose
Project Charter	Formal authorization; captures high-level scope, objectives, budget, authority.
WBS	Breaks work into manageable units; feeds cost and schedule estimates.
Schedule (PERT/CPM, Gantt)	Visualizes activity sequencing, durations, and critical path.
Risk Register	Lists threats and opportunities with mitigation plans.

- **Critical Path Analysis (CPA)** identifies the longest dependent activity chain; any delay on this path extends the overall project duration.
- **Float types** clarify flexibility:
 - **Total Float** – delay allowable without impacting project finish.
 - **Free Float** – delay allowable without affecting immediate successors.
 - **Independent Float** – delay allowable without affecting either predecessors or successors (worst-case predecessor, best-case successor).

Managing Uncertainty with PERT

- **PERT** treats activity durations probabilistically, using **three-point estimates**:

Estimate	Symbol	Typical meaning
Optimistic	A	Best-case, no delays.
Most Likely	M	Normal operating conditions.
Pessimistic	B	Worst-case, many delays.

- Expected duration for an activity:

$$TE = \frac{A+4M+B}{6}$$

- Variance for the activity:

$$TV = \left(\frac{B-A}{6} \right)^2$$

- The **project duration** derived from PERT corresponds to the **median** (50 % probability) of the finish-time distribution, giving a realistic expectation of on-time completion.

Management Roles & Organizational Structures

- **Project manager** balances internal duties (WBS creation, schedule maintenance, risk monitoring, quality assurance) with external duties (status reporting, stakeholder liaison, resource negotiation).
- **Managerial roles** (Mintzberg) most relevant to projects:
 - **Leader** – motivates and guides the team.
 - **Liaison** – builds external contacts.
 - **Monitor** – gathers information on progress and risks.
 - **Entrepreneur** – initiates improvements or new initiatives.
 - **Disturbance Handler** – resolves conflicts and unexpected changes.
- **Organizational structures** determine reporting lines and resource allocation:

Structure	Characteristics
Functional	Teams organized by specialty; projects flow through departmental pipelines.
Project-based	Participants assigned to a single project; disbanded after delivery.
Matrix	Dual reporting (functional manager + project manager); blends expertise with flexibility.

Synthesizing the Pieces

- The **WBS** feeds directly into both **CPA** (by defining activity relationships) and **PERT** (by supplying the three-point estimates for each activity).
- **Float calculations** derived from CPA complement PERT's probabilistic view, offering both deterministic slack and stochastic risk insight.
- **Management theories** (Fayol's principles and Mintzberg's roles) provide the human-behavior framework that underpins successful execution of the technical planning artifacts.

Together, these concepts form a cohesive ecosystem: a clear project definition, structured decomposition, rigorous schedule analysis, probabilistic risk handling, and effective leadership—all essential for delivering unique, time-bound outcomes within defined constraints.